

Lab Assignment IV

Database Management Systems

Technical Stack Selection: Online Course Management Platform

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1 Introduction

As per the requirements for the Online Course Management Platform , this project involves the design of a complete web-based information system. The architecture consists of a backend database, a connectivity server, front-end scripts, and a form-based web interface. This report justifies the selected technology stack based on its ability to satisfy the design objectives and functional requirements.

2 Proposed Technology Stack

The following technologies have been selected for the implementation of the platform:

- **Frontend:** HTML5, CSS3, and JavaScript
- **Connectivity Server (Backend):** Python with Flask
- **Database Server:** PostgreSQL

3 Requirement Mapping and Justification

3.1 Frontend: HTML, CSS, and JavaScript

The assignment requires an "easy to use interface" and a "form-based web interface".

- **HTML5 & CSS3:** These will be used to build the "entry page" and the structural layout for the various menus. They ensure the interface is intuitive for managing courses, programs, and user types.
- **JavaScript (Front-end Scripts):** As "scripts are compulsory", JavaScript will handle the dynamic aspects of the frontend. Specifically, it will facilitate the requirement for students to "view and search various courses".

3.2 Connectivity Server: Flask (Python)

The system requires a "connectivity server" to bridge the interface and the database.

- **Functional Logic:** Flask will manage the login processes and session handling for the various users: System Administrator, Instructors, Students, and Data Analysts.
- **Dynamic Content:** Flask allows the system to "display various menu based on category of the user" and enables instructors to "add content to a course".

3.3 Database Server: PostgreSQL

The use of a database is "compulsory" for this system to manage all data related to courses, teachers, and students.

- **Data Management:** PostgreSQL will store information for the System Administrator to manage teacher assignments and student records.
- **Statistical Analysis:** The relational nature of PostgreSQL is essential to "generate statistics about the course enrolments" for the Data Analyst role.

4 Summary of Technology Application

The following table summarizes how each technology satisfies the specific requirements mentioned in the assignment:

Component	Technology	Assignment Requirement Fulfilled
Frontend	HTML/CSS	Form-based interface; Easy to use interface
Front-end Scripts	JavaScript	Course searching; Compulsory scripts
Connectivity Server	Flask	Login handling; Category-based menus
Backend Database	PostgreSQL	Data management; Enrollment statistics

5 Conclusion

The combination of HTML/CSS/JS, Flask, and PostgreSQL provides a complete solution that adheres to the mandatory use of scripts and databases. This stack ensures all functional requirements—from user registration to statistical data analysis—are met through a robust, web-based platform

